MSG No. 128.406 $P4'/m'nc'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4'/m'..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,5,6,9,10,13,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,4,7,8,11,12,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `4'/m'..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,5,6,9,10,13,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,4,7,8,11,12,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,13,14]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,4,15,16]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,8,11,12]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,10,11,12]`
4	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[13,14,15,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `4'..`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,9,10]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[3,4,11,12]`
3	$[0, 0, -z]$	`[5,6,13,14]`
4	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[7,8,15,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,2]`
2	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	`[3,4]`
3	$\left[\frac{1}{2}, 0, -z\right]$	`[5,6]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[7,8]
5	$\left[\frac{1}{2}, 0, z\right]$	[9,10]
6	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[11,12]
7	$\left[0, \frac{1}{2}, -z\right]$	[13,14]
8	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[15,16]

Table 7: Wyckoff site: 8g, site symmetry: ..2

No.	position	mapping
1	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	[1,3]
2	$\left[-x, \frac{1}{2}-x, \frac{1}{4}\right]$	[2,4]
3	$\left[x+\frac{1}{2}, -x, \frac{3}{4}\right]$	[5,8]
4	$\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$	[6,7]
5	$\left[\frac{1}{2}-x, x, \frac{1}{4}\right]$	[9,12]
6	$\left[x+\frac{1}{2}, -x, \frac{1}{4}\right]$	[10,11]
7	$\left[-x, \frac{1}{2}-x, \frac{3}{4}\right]$	[13,15]
8	$\left[x, x+\frac{1}{2}, \frac{3}{4}\right]$	[14,16]

Table 8: Wyckoff site: 8h, site symmetry: m'..

No.	position	mapping
1	$[x, y, 0]$	[1,14]
2	$[-x, -y, 0]$	[2,13]
3	$\left[y+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[3,16]
4	$\left[\frac{1}{2}-y, \frac{1}{2}-x, \frac{1}{2}\right]$	[4,15]
5	$[y, -x, 0]$	[5,10]
6	$[-y, x, 0]$	[6,9]
7	$\left[\frac{1}{2}-x, y+\frac{1}{2}, \frac{1}{2}\right]$	[7,12]
8	$\left[x+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	[8,11]

Table 9: Wyckoff site: 16i, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[-x, -y, z]$	[2]
3	$\left[y+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-z\right]$	[3]
4	$\left[\frac{1}{2}-y, \frac{1}{2}-x, \frac{1}{2}-z\right]$	[4]
5	$[y, -x, -z]$	[5]
6	$[-y, x, -z]$	[6]
7	$\left[\frac{1}{2}-x, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[7]
8	$\left[x+\frac{1}{2}, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[8]
9	$[-y, x, z]$	[9]
10	$[y, -x, z]$	[10]

continued ...

Table 9

No.	position	mapping
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[12]
13	$[-x,\ -y,\ -z]$	[13]
14	$[x,\ y,\ -z]$	[14]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16]